

LINK TO REPOSITORY:

<https://github.com/Ojas2011/Impact-of-social-media-use-and-gaming-on-daily-mood.git>

Impact of social media use and gaming on daily mood

Introduction

Through this project, our group aimed to answer the research question: How accurately can hours of social media and gaming usage, combined with self-reported psychological responses, predict the day-to-day mental health outcomes (daily mood) of individuals?

The reason this project is important and connects with society more is because of the fact that today's society (especially today's youth) spend a lot of time playing video games as well as a lot of time on social media, we aimed to see how their lifestyle was impacted (whether they were being helped/hurt and how and how much) from a sample of more than 70 people that we were able to collect.

When it comes to people's mental health there are many things that can contribute to why someone feels a certain way as well as what could have led to different people of the same communities and groups having different kinds of responses to mental health related questions. Through this project, our group decided to see how the two factors (social media use and time spent gaming) were impacting people's mental health and their overall wellbeing.

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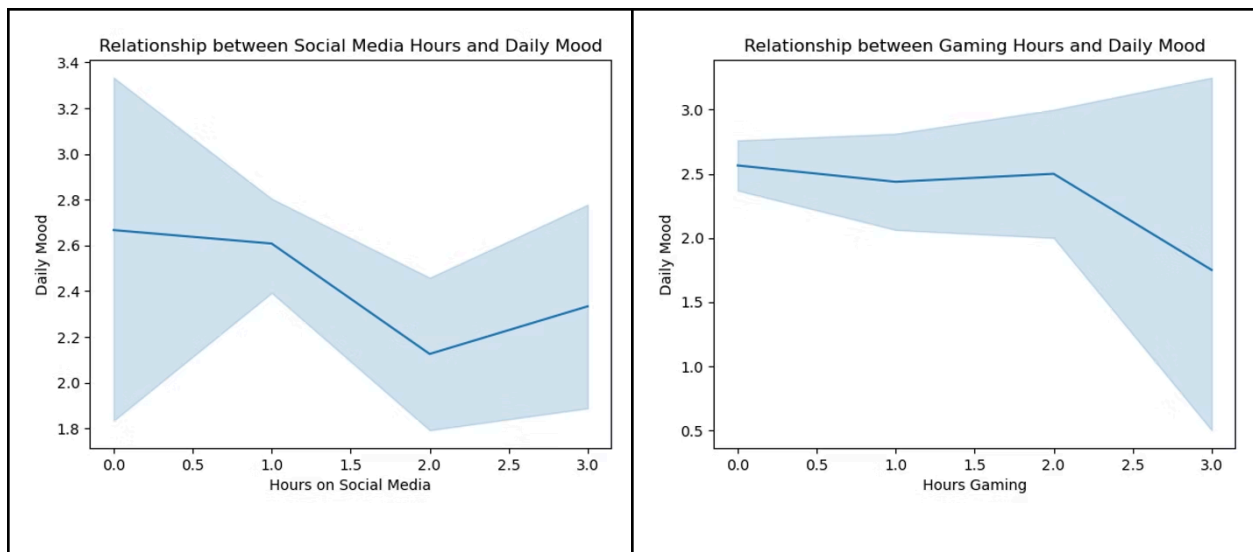
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Data

To get the data, we sent out a survey to multiple people at UNC Charlotte and got over a 100 responses regarding how sleep, stress levels, mood, etc were impacted. We decided to make Daily Mood our target variable since it correlated with the impact of gaming and social media.

After receiving more than a 100 responses, we began cleaning our data and getting 70 usable rows from the 100 raw responses as well as 11 features after cleaning, as well as with the 80/20 train/test split getting 56 training samples and 14 test samples.

When it came to the preprocessing steps, we dropped 17 irrelevant / free-text columns to clean data as well as many empty responses (since many questions were not marked as required), followed by ordinal encoding for social media and video game hours, as well as median imputation (SimpleImputer) for median values), followed by target binned [0-1 {bad}], [2 {neutral}], [3-4 {good}]. And finally removing erroneous 'yes' values from social media hours.

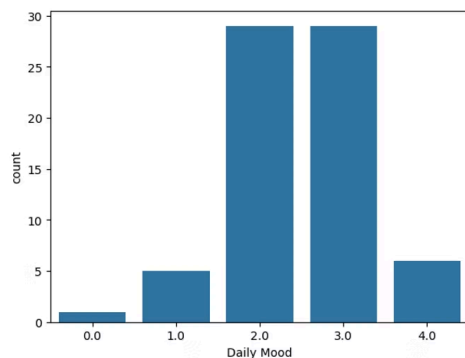


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The above graph shows that there is not really a strong relation between hours on social media and daily mood with there being a very weak linear correlation.

The above graph shows that there is a relatively more stable measurement of daily mood in the beginning and as you hit two hours of gaming it declines, it's still not as good of a predictor but compared to hours on social media hours gaming does tell a little more of a story.



The bar graph shown above separates the values in 'Daily Mood' into numerical clusters on a scale of 0-4.

As shown here most respondents were in the 2.0-3.0 range which is moderate to positive (though not as positive).

Models Used

1. KNN (K Nearest Neighbors): Showed similarities in trends in responses and behaviors using Euclidean distance, StandardScaler. k=1-20 tuned; k=6 optimal with best train-test gap (58.9% to 64.3%). The score for this model was **64.3%**.

2. Decision Tree (Feature Selection): Helped figure out which features were having a bigger impact on our results that we were getting and were showing that quality sleep time and mental health had been influenced. The score for this model was **57.1%**.

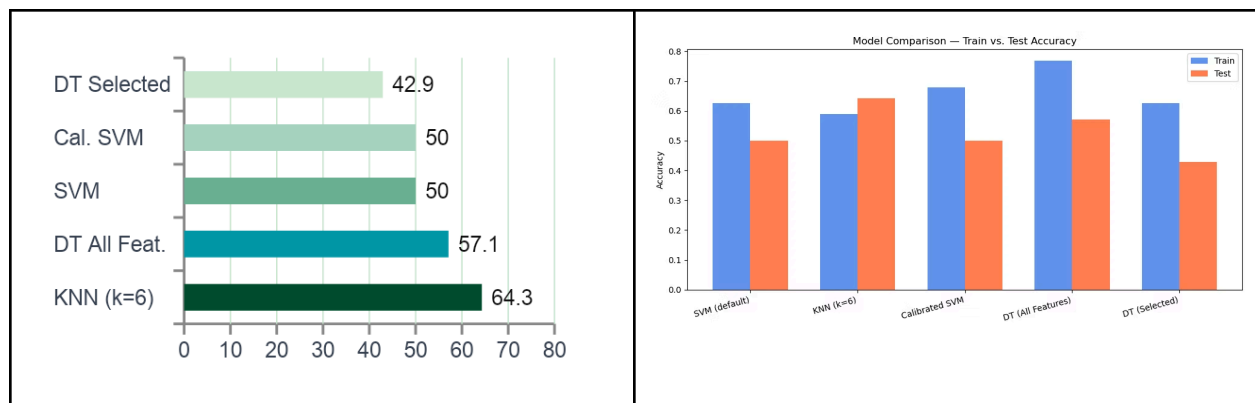
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3. SVM (Support Vector Machine): Helped separate two different classes, involved the RBF kernel, class_weight being balanced, finding max-margin as well as hyperplane minimizing $\|w\|^2$ s.t. $Y_i (w \cdot X_i + b) \geq 1$. The score for this model was **50%**.

- Something else worth mentioning was getting a linear regression or R^2 of -0.24 which is relatively low but given the kind of dataset we were dealing with it was relatively better.

Results



KNN Classifier has the best test accuracy at 64.3% which while is something definitely that can be improved, given the lack of data/responses we had it was definitely a good accuracy.

Something else that was found to be important was that how much time spent gaming/on social media, while having an impact was not the only thing that was changing people's moods/sleep, it also had to do with how they were using the technology such as social media, since it was also found that trying to stop gaming/spending time on social media was causing respondents to want to do it even more.

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Psychological side-effects were also key players in this situation such as: anxiety, sleep disruption, addiction & loneliness outranking raw screen-time hours as predictors of daily mood.

Importance (Revisited)

The reason for originally choosing this topic has to do with many things we learnt prior to starting the project, including but not limited to things such as the fact that as per Engadget, “Around 9% of teenagers have reported being negatively impacted by social media or gaming... TikTok alone has been linked to sleep disruption in over 37% of teen users.”, the difference between our inspiration for the project and our project itself is the amount of variables we got was significantly less, as well as the fact that because most respondents were within Charlotte NC itself the data could only answer about a certain state or a certain city.

It is important to understand what causes problems with mood and sleep and if social media/video games have anything to do with it, the model(s) used in this project can help with such predictions when needing to understand where things are going wrong and where to start to make them right (though not completely but definitely much more the larger and broader the dataset).

Technical Findings

Social Media and Video Games themselves are not the problem, even sites such as Psychology Today say that “social media can actually build a community and spread vital and important information rapidly, as we saw during the COVID-19 pandemic.”

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It is also worth noting that as per Open Textbooks Research “gaming can actually sharpen problem-solving skills and decision-making. The issue is not the technology itself. It is more about how we interact with it.”

The problem is the overuse of both social media and spending more than enough time playing video games when it’s not exactly going to help you, for example if you are going to use social media for the sole purpose of entertainment then after a certain point it will be harmful to your brain but if it is educational content being watched then while after a certain point it still could be bad the damage won’t be as bad as the former.

Limitations & Ethics

It was a small sample we took to make conclusions from and therefore results may not generalize nationally. It is also worth noting that the responses are self-reported rather than from an expert/spectator. What else is important to note is a class imbalance the ML models failed to predict even though it was one ‘bad’ sample in the test set.

This project is also not meant to make conclusions for doctors/psychologists since the data gathered may not be accurate and me and my team are not experts in the field, moreover, the purpose of this survey was for educational purposes rather than clinical purposes.

Everyone’s responses to the survey were anonymous meaning there was no personally identifiable information, as well as the fact that everyone who filled the survey out voluntarily did so and selected the answers they did voluntarily.

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Conclusion

When working on a project like the one worked on here in the future, something to take into account will be having a larger dataset as well as using different models maybe if needed (though likelihood seems low for that to be needed), also need to see statistical values such as linear regression and how far it is from very low percentages, preferably higher than -0.5 or 0.5 for better accuracy, same with the models, also having more qualitative data to be actually evaluated next time, overall having more and doing more from it next time.

AI Transparency

Claude (Anthropic, claude-sonnet-4-6) as well as Gemini, ChatGPT and github copilot (embedded in VS code) were used for help with this project when it came to AI use.

References

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